

ISyE6669 Homework 7

Fall 2025

1 Week 7

1. Geometry of LP

- (a) Let P be a triangle in a plane with three extreme points $A = [1, 1]^\top, B = [2, 4]^\top, C = [4, 2]^\top$. Let $x = [2, 2]^\top$ be a point in the triangle P . Express x as a convex combination of the three extreme points A, B, C . You should write $x = \lambda_1 A + \lambda_2 B + \lambda_3 C$, where $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and all nonnegative. You need to give the numerical values of $\lambda_1, \lambda_2, \lambda_3$.

Solution: We need to solve for $\lambda_1, \lambda_2, \lambda_3$ in the equation $x = \lambda_1 A + \lambda_2 B + \lambda_3 C$.

Substituting the given values:

$$\begin{bmatrix} 2 \\ 2 \end{bmatrix} = \lambda_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \lambda_2 \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \lambda_3 \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

This gives us the system of three equations:

$$\lambda_1 + 2\lambda_2 + 4\lambda_3 = 2$$

$$\lambda_1 + 4\lambda_2 + 2\lambda_3 = 2$$

$$\lambda_1 + \lambda_2 + \lambda_3 = 1$$

Solving this system of linear equations:

$$\lambda_1 = \frac{1}{2}, \quad \lambda_2 = \frac{1}{4}, \quad \lambda_3 = \frac{1}{4}$$

Therefore, $x = \frac{1}{2}A + \frac{1}{4}B + \frac{1}{4}C$.

- (b) Consider the following polyhedron P defined as

$$P = \{(x, y) \in \mathbb{R}^2 : x + 2y \geq 3, x - y \leq 1, y \geq 1, x \leq -1\}.$$

Find all the extreme points and extreme rays of P .

Solution: Please refer to the accompanying figure for the graphical representation of the polyhedron.

Extreme Points: The polyhedron P has exactly one extreme point: $(-1, 2)$.

Extreme Rays: The polyhedron P has two extreme rays:

- Direction vector $(0, 1)$

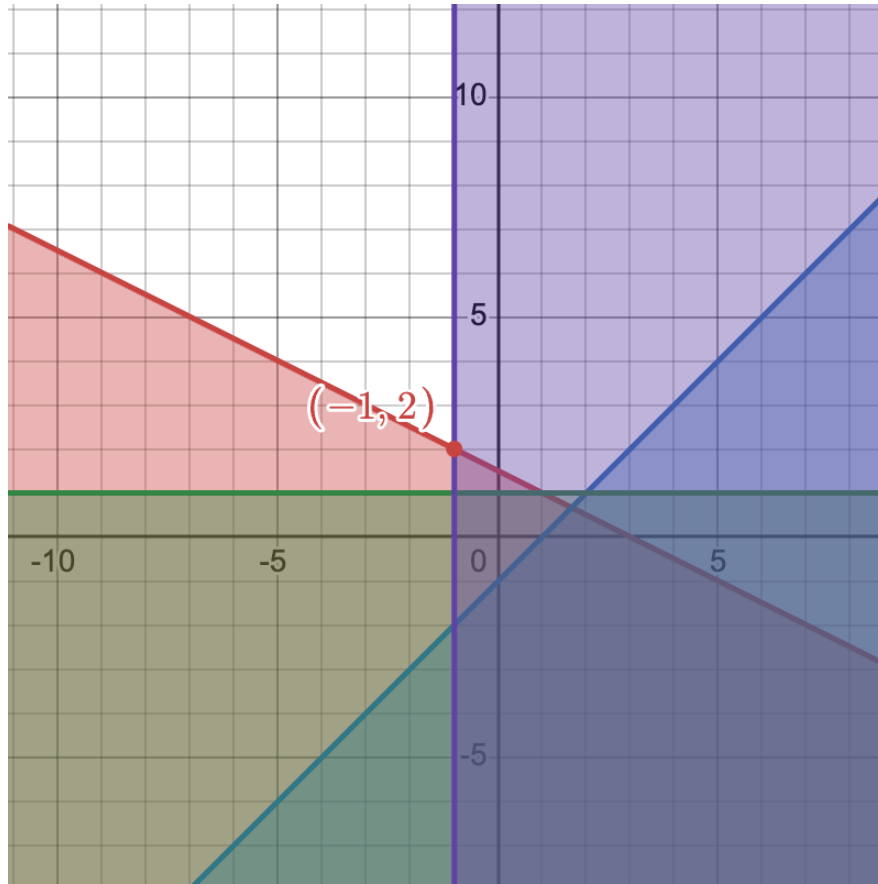


Figure 1: Polyhedron P with extreme point and extreme rays

- Direction vector $(-2, 1)$
- (c) Remember that a polytope is a bounded polyhedron. Give an example of a polytope in dimension n defined by exactly $2n$ inequalities, such that removing any inequality describing the polytope makes the resulting polyhedron unbounded. Hint: First think of an example in $\mathbb{R}^2$ and then generalize it.

Solution: Consider the unit hypercube in $\mathbb{R}^n$ defined by the following $2n$ inequalities:

$$x_i \geq 0 \quad \text{and} \quad x_i \leq 1 \quad \text{for } i = 1, 2, \dots, n$$

This polytope is bounded since all coordinates are constrained between 0 and 1. However, removing any single inequality makes the polyhedron unbounded:

- Removing $x_i \geq 0$ allows the i -th coordinate to extend to $-\infty$
- Removing $x_i \leq 1$ allows the i -th coordinate to extend to $+\infty$

Therefore, the unit hypercube provides the desired example of a polytope in dimension n with exactly $2n$ inequalities where removing any inequality results in an unbounded polyhedron.

By the way, an interesting fact is that examples of the above kind do not occur if the number of inequalities is more than $2n$. For example if a polytope is in dimension n and is defined by $2n + 1$ inequalities, then there always exists one inequality such that removing this inequality leaves the resulting (possibly larger) polyhedron bounded. I encourage you to experiment in $n = 2$ dimension to see this fact!

2. Reformulation as linear program

(a) Consider the following linear program:

$$\begin{array}{ll}\max & x + 2y \\ \text{s.t.} & x + y \leq 4, \\ & 2x - y \geq -2, \\ & 2x + y \leq 6, \\ & x \leq 0.\end{array}$$

Convert the above LP to a standard form LP.

Solution: To convert to standard form, we introduce the following variable transformations:

- Let $x' = -x$ (since $x \leq 0$, we have $x' \geq 0$)
- Let $y = y^+ - y^-$ where $y^+, y^- \geq 0$ (to handle unrestricted y)
- Add slack variables $s_1, s_2, s_3 \geq 0$

The standard form LP becomes:

$$\begin{array}{ll}\min & -x' - 2y^+ + 2y^- \\ \text{s.t.} & -x' + y^+ - y^- + s_1 = 4 \\ & -2x' - y^+ + y^- - s_2 = -2 \\ & -2x' + y^+ - y^- + s_3 = 6 \\ & x', y^+, y^-, s_1, s_2, s_3 \geq 0\end{array}$$

(b) Alexis Cornby makes her living buying and selling corn. On January 1, she has 50 tons of corn and \$1000. On the first day of each month Alexis can buy corn at the following prices per ton: January \$300; February, \$350; March, \$400; April, \$500. On the last day of each month, Alexis can sell corn at following prices per ton: January, \$250; February, \$400; March, \$350; April, \$550. Alexis stores her corn in a warehouse that can hold at most 100 tons of corn. She must be able to pay cash for all corn at the time of purchase. Write a linear programming model that can be solved to determine how Alexis can maximize her cash on hand at the end of April.

Solution: Let us define the following variables:

- X_{iB} = tons of corn bought in month i (for $i = 1, 2, 3, 4$)
- X_{iS} = tons of corn sold in month i (for $i = 1, 2, 3, 4$)
- $X_0 = 50$ = initial corn inventory
- $P_0 = 1000$ = initial cash

Objective Function: Maximize cash on hand at the end of April:

$$\begin{aligned} \max \quad & P_0 - 300X_{1B} + 250X_{1S} - 350X_{2B} + 400X_{2S} \\ & - 400X_{3B} + 350X_{3S} - 500X_{4B} + 550X_{4S} \end{aligned}$$

Constraints:

1. Inventory constraints (non-negative inventory):

$$\begin{aligned} X_0 + X_{1B} - X_{1S} &\geq 0 \\ X_0 + X_{1B} - X_{1S} + X_{2B} - X_{2S} &\geq 0 \\ X_0 + X_{1B} - X_{1S} + X_{2B} - X_{2S} + X_{3B} - X_{3S} &\geq 0 \\ X_0 + X_{1B} - X_{1S} + X_{2B} - X_{2S} + X_{3B} - X_{3S} + X_{4B} - X_{4S} &\geq 0 \end{aligned}$$

2. Cash constraints (non-negative cash):

$$\begin{aligned} P_0 - 300X_{1B} &\geq 0 \\ P_0 - 300X_{1B} + 250X_{1S} - 350X_{2B} &\geq 0 \\ P_0 - 300X_{1B} + 250X_{1S} - 350X_{2B} + 400X_{2S} - 400X_{3B} &\geq 0 \\ P_0 - 300X_{1B} + 250X_{1S} - 350X_{2B} + 400X_{2S} - 400X_{3B} + 350X_{3S} - 500X_{4B} &\geq 0 \end{aligned}$$

3. Storage capacity constraints (at most 100 tons):

$$\begin{aligned} X_0 + X_{1B} - X_{1S} &\leq 100 \\ X_0 + X_{1B} - X_{1S} + X_{2B} - X_{2S} &\leq 100 \\ X_0 + X_{1B} - X_{1S} + X_{2B} - X_{2S} + X_{3B} - X_{3S} &\leq 100 \\ X_0 + X_{1B} - X_{1S} + X_{2B} - X_{2S} + X_{3B} - X_{3S} + X_{4B} - X_{4S} &\leq 100 \end{aligned}$$

4. Non-negativity constraints:

$$X_{1B}, X_{2B}, X_{3B}, X_{4B}, X_{1S}, X_{2S}, X_{3S}, X_{4S} \geq 0$$

3. Basic Solution and Basic Feasible Solution

(a) Consider the following linear program:

$$\begin{aligned} \min \quad & -x_1 - 2x_2 \\ \text{s.t.} \quad & x_1 - x_2 \leq 1 \\ & 2x_1 - x_2 \geq -1 \\ & 2x_1 + x_2 \leq 5 \\ & x_1 \geq 0, \quad x_2 \geq 0. \end{aligned}$$

Graph the constraints of this linear program, and indicate the feasible region.

Solution: Please refer to the accompanying figure for the graphical representation of the constraints and feasible region.

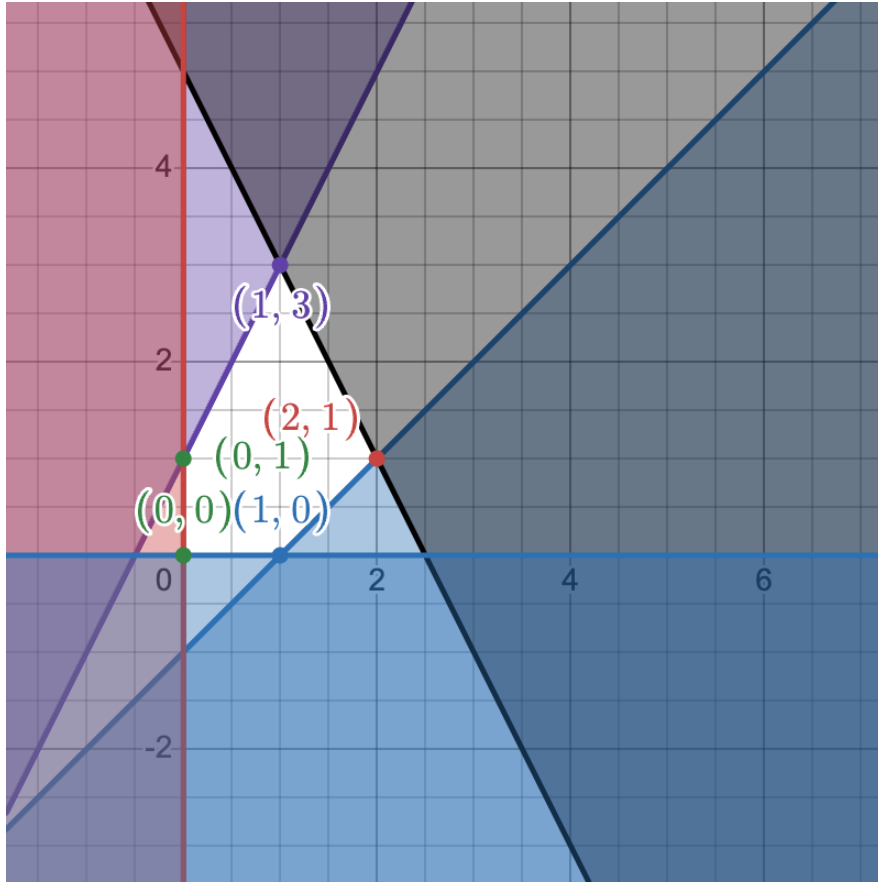


Figure 2: Linear program constraints and feasible region

(b) Transform it into a standard form LP.

Denote $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5]^\top$ as the vector of variables, and use the standard form notation:

$$\begin{aligned} \min \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{Ax} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}, \end{aligned}$$

specify $\mathbf{c}$, $\mathbf{A}$, $\mathbf{b}$ for the above problem.

Solution: We introduce slack variables x_3, x_4, x_5 to convert inequalities to equalities. The standard form LP becomes:

$$\begin{aligned} \min \quad & -x_1 - 2x_2 \\ \text{s.t.} \quad & x_1 - x_2 + x_3 = 1 \\ & -2x_1 + x_2 + x_4 = 1 \\ & 2x_1 + x_2 + x_5 = 5 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0 \end{aligned}$$

Therefore, we have:

$$\mathbf{c} = \begin{bmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & -1 & 1 & 0 & 0 \\ -2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$$

- (c) Use the procedure discussed in lecture to find *all* basic solutions. For each basic solution, specify the basis matrix, the basic variables, the non-basic variables, and the associated cost coefficients for each part. For example, for the following basis matrix

$$\mathbf{B} = [\mathbf{A}_3, \mathbf{A}_4, \mathbf{A}_5] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

where $\mathbf{A}_i$ is the i -th column of $\mathbf{A}$, the corresponding basic variables, non-basic variables, and cost coefficients are

$$\mathbf{x}_B = \begin{bmatrix} x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 5 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -1 \\ -2 \end{bmatrix}.$$

Specify all basic solutions in this way. [Hint: The nonbasic variables are always zero. So the key is to compute the basic variables. It will require inverting the basis matrix $\mathbf{B}$. You can use Python to compute the inverse of a matrix. Scipy has very fast linear algebra packages. You only need to invoke “from scipy import linalg” and use “linalg.inv(A)” to invert a matrix A .]

Solution: There are $\binom{5}{3} = 10$ possible combinations of basic variables. Here are all basic solutions:

1. Basic variables: x_1, x_2, x_3 ; **Non-basic variables:** x_4, x_5

$$\mathbf{B} = \begin{bmatrix} 1 & -1 & 1 \\ -2 & 1 & 0 \\ 2 & 1 & 0 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

2. Basic variables: x_1, x_2, x_4 ; **Non-basic variables:** x_3, x_5

$$\mathbf{B} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

3. Basic variables: x_1, x_2, x_5 ; **Non-basic variables:** x_3, x_4

$$\mathbf{B} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 1 & 0 \\ 2 & 1 & 1 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} -2 \\ -3 \\ 12 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

4. **Basic variables:** x_1, x_3, x_4 ; **Non-basic variables:** x_2, x_5

$$B = \begin{bmatrix} 1 & 1 & 0 \\ -2 & 0 & 1 \\ 2 & 0 & 0 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} 2.5 \\ -1.5 \\ 6 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

5. **Basic variables:** x_1, x_3, x_5 ; **Non-basic variables:** x_2, x_4

$$B = \begin{bmatrix} 1 & 1 & 0 \\ -2 & 0 & 0 \\ 2 & 0 & 1 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} -0.5 \\ 1.5 \\ 6 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

6. **Basic variables:** x_1, x_4, x_5 ; **Non-basic variables:** x_2, x_3

$$B = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

7. **Basic variables:** x_2, x_3, x_4 ; **Non-basic variables:** x_1, x_5

$$B = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} 5 \\ 6 \\ -4 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

8. **Basic variables:** x_2, x_3, x_5 ; **Non-basic variables:** x_1, x_4

$$B = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

9. **Basic variables:** x_2, x_4, x_5 ; **Non-basic variables:** x_1, x_3

$$B = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

10. **Basic variables:** x_3, x_4, x_5 ; **Non-basic variables:** x_1, x_2

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

- (d) Among all the basic solutions you found, which basic solutions are feasible, thus are basic feasible solutions? Which basic solutions are infeasible? Locate each basic solution on the graph you drew in part (a). [Hint: You only need to look at the (x_1, x_2) part of each basic solution.] What is the optimal solution?

Solution: Based on the accompanying figures showing each basic solution plotted on the feasible region:

Feasible Basic Solutions (Basic Feasible Solutions):

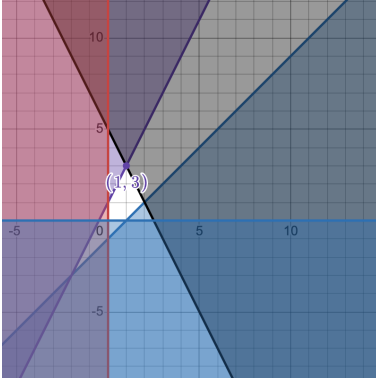


Figure 3: Solution 1: Feasible

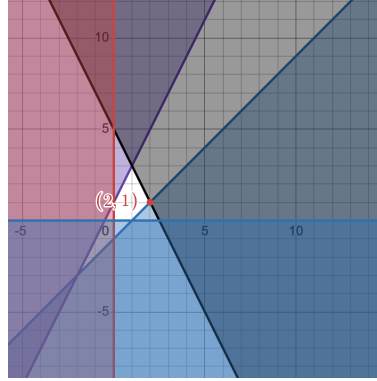


Figure 4: Solution 2: Feasible

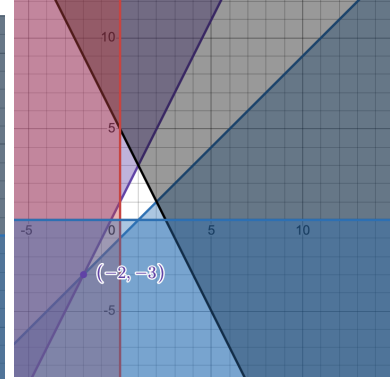


Figure 5: Solution 3: Infeasible

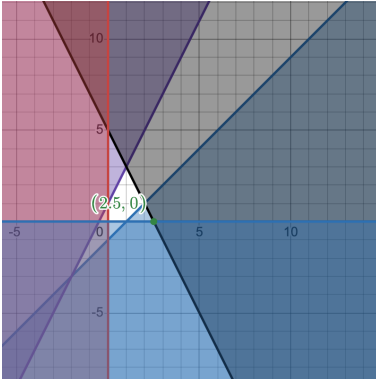


Figure 6: Solution 4: Infeasible

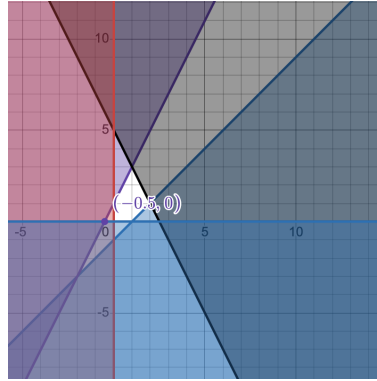


Figure 7: Solution 5: Infeasible

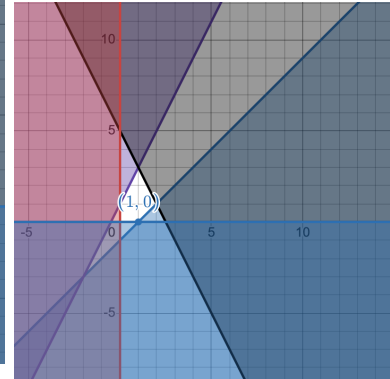


Figure 8: Solution 6: Feasible

- Solution 1: $(x_1, x_2) = (1, 3)$ with objective value -7
- Solution 2: $(x_1, x_2) = (2, 1)$ with objective value -4
- Solution 6: $(x_1, x_2) = (1, 0)$ with objective value -1
- Solution 8: $(x_1, x_2) = (0, 1)$ with objective value -2
- Solution 10: $(x_1, x_2) = (0, 0)$ with objective value 0

Infeasible Basic Solutions:

- Solution 3: $(x_1, x_2) = (-2, -3)$ (violates non-negativity constraints)
- Solution 4: $(x_1, x_2) = (2.5, 0)$ (outside feasible region)
- Solution 5: $(x_1, x_2) = (-0.5, 0)$ (violates non-negativity constraint)
- Solution 7: $(x_1, x_2) = (0, 5)$ (outside feasible region)
- Solution 9: $(x_1, x_2) = (0, -1)$ (violates non-negativity constraint)

Optimal Solution: The optimal solution is **Solution 1** with $(x_1, x_2) = (1, 3)$ and optimal objective value -7 . This corresponds to the corner point of the feasible region that minimizes the objective function $-x_1 - 2x_2$.

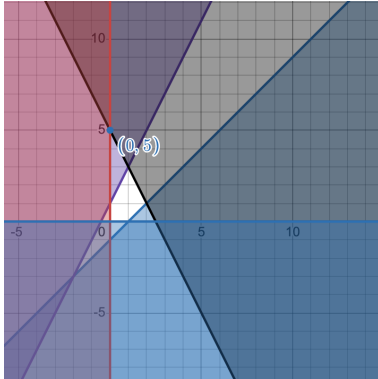


Figure 9: Solution 7: Infeasible

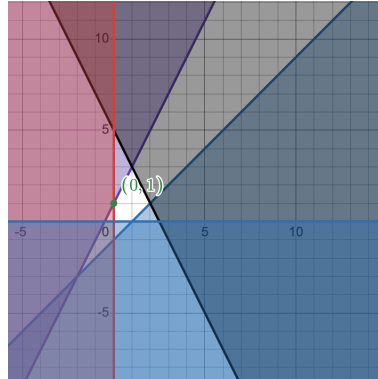


Figure 10: Solution 8: Feasible

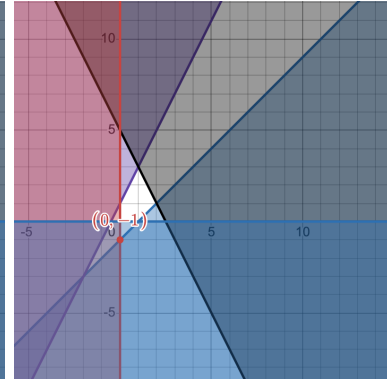


Figure 11: Solution 9: Infeasible

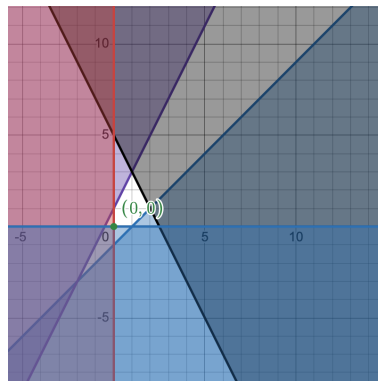


Figure 12: Solution 10: Feasible